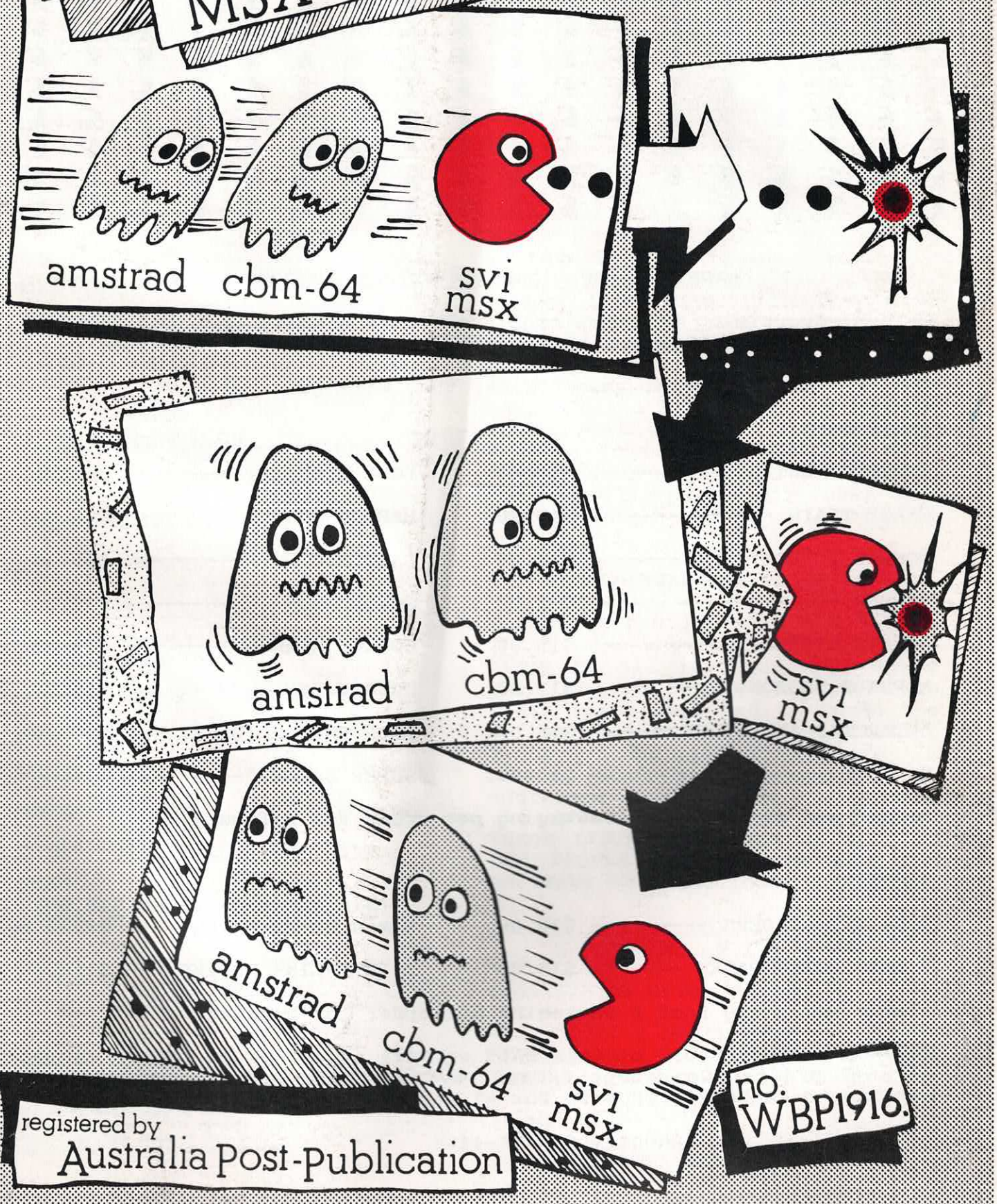
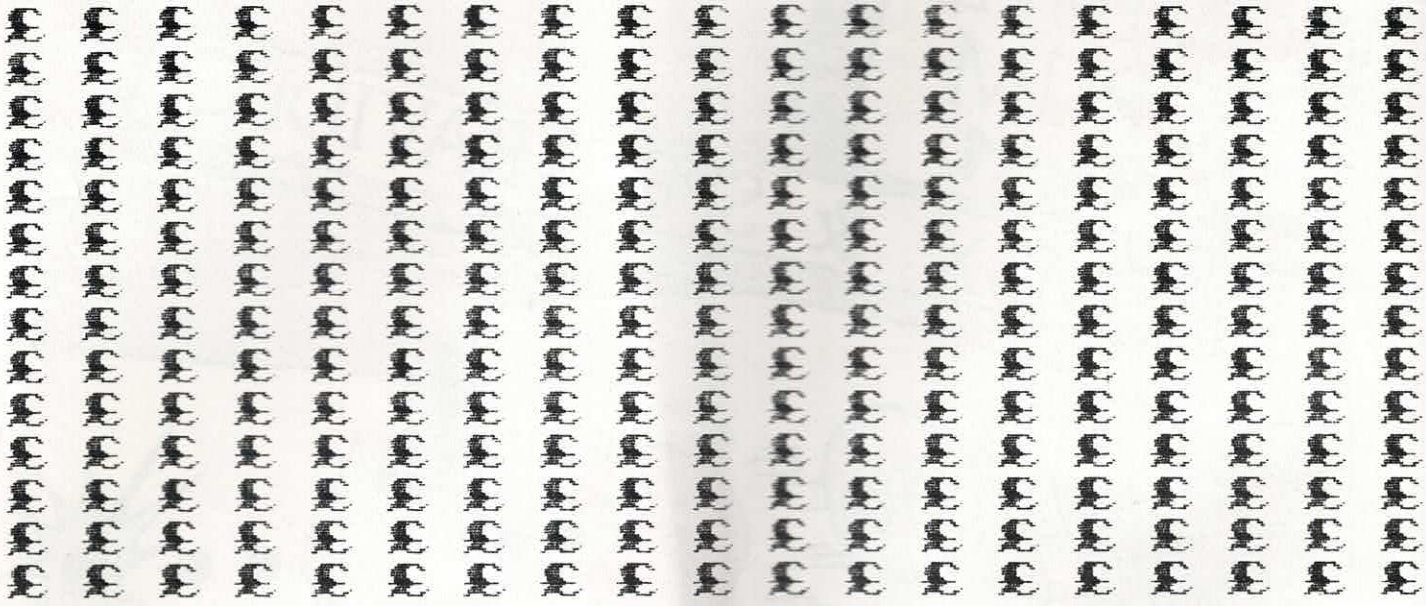
COMPUTER FORUM

VOLUME. 2 NO. 11

MSX & SPECTRAVIDEO



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PRINTED BY - Snap Printing

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EDITORIAL

--- WAITING ---

I suppose you have heard of the mail strike in England!!, well we have had no word from them at all, so it looks like we will have to wait until the next issue to find out about NEMESIS & NEMESIS2. We had a small response to my little Editorial in the last issue about the games.

----- PEN PALS -----

The name of our magazine has been put into a Japanese magazine called - MSX MAGAZINE. I don't know what the ad said, but about 3 weeks ago I started getting letters from people in Japan wanting to become **penpals** with anyone from Australia. They wish to swap ideas and programs if possible except for one minor detail 99% of them have MSX2 but it is possible that they may also have MSX1. I have about 70 letters so far, if you are interested in writing to them please send me a **STAMPED SELF ADDRESSED ENVELOPE** and I will put their name and address in it and send it back to you.

------ SOFTWARE SWAPPING ------

I know in the last issue I said I would send out all the software lists to all involved, but since then 2 more people have sent in their lists so I had to re-do the software lists. If you would like a copy of the list of software please send in a stamped self addressed envelope and I will send you a list of the software. **THESE LISTS WILL ONLY BE SENT TO THE PEOPLE WHO HAVE ENTERED THE SOFTWARE SWAPPING IDEA.**

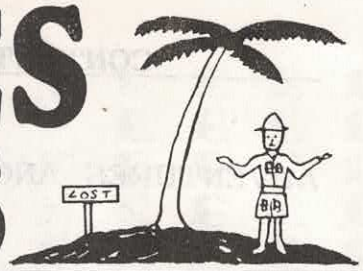
C O V E R W I N N E R

This issues cover winner is -

--- ????? I DON'T KNOW ---

Whoever sent in this cover please let us know your Name & Address.

ADVENTURES ANONYMOUS



WITH YOUR GUIDE ALAN CRAWFORD

Welcome once again to AA. As I mentioned a short time ago I was expecting a couple of adventure games to be sent to me from LPG software in Tasmania, I have in fact now recieved the games but unfortunetly I was only sent part one of JOURNEY TO THE CENTRE OF THE EARTH. As this adventure is a multi part game on the same tape I feel that I cannot do the game justice by reviewing only one part of the game except to say that it looked good on the screen but part one was just to simple to complete (it took about half an hour). It does have a lot of commands that where not used in part one, so it is more than likely that the game would improve as it goes on. It does appear however to be a good game for the beginner.

As promised last month its time to take a look at the adventures from LEVEL 9, because of the size of these adventures it will be neccesary to split them up over a few issues.

LORDS OF TIME.

SCENARIO.

LORDS OF TIME as the name suggests is set in many different time zones, nine in fact, they range from the ice age to the future. The game revolves around a clock which you use to travel from one time zone to another collecting objects to eventually defeat the evil timelords who are meddling with earths history.

Each of the nine time zones should be tackled in numerical order to ensure you collect all the objects which comprise of nine ingredients (one in each zone) and eighteen treasures which are scattered throughout the game. You can return to the clock from the first eight time zones only. Certain objects that you find will be needed in a different time zone, so if you find no use for a certain object do not discard it. To help you along I will list all special objects and which time zone they are in.

INGREDIENTS.	T/Z	TREASURES.	T/Z	TREASURES.	T/Z
Diamond Teardrop	1.	Hourglass (start)		Lute	6.
Ivory Tusk	2.	Lodestone	1.	Crown	6.
Dinosaur Egg	3.	Metronome	1.	Drinking Horn	6.
Dragons Wing	4.	Jewellery case	1.	Oriental Rug	6.
Olive Branch	5.	Candelabra	2.	Emerald	7.
Jesters Cap	6.	Onyz Figurine	3.	Phial	7.
Silicon Chip	7.	Stone Pot	3.	Ruby	7.
Golden Buckle	8.	The Wheel	3.	Fallen Star	7.
Box (evil eye)	9.	Jewelled Coffer	4.	Cloak	9.

All nine ingredients are to be collected and used in time zone nine. As you get them, drop them in the clock. Some of the treasures will be needed to help you accomplish this.

START.

The living room is where you start and the HOURGLASS is the eirst treasure you find. There is also a picture here which you should look at. From here go NORTH to the large cold room to find matches, candle and the clock. The door to the clock can be opened by winding the clock, you only need to do this once, after that you only need to 'open door'. Once inside the clock you will find cogs and a pendulum, these are used to operate the time machine. The cogs are numbered 1 to 9 and correspond to the time zones. Select the time zone you want by turning the cog, eg. turn cog 1. then push pendulum to set the clock in motion. Then open door and go north to leave the clock. This sequence should be used every time you want to travel to a different time zone. Just change the cog number to the corresponding time zone.

TIME ZONE 1.

Leave the clock and head north to the roadworks to get the pick and the planks. Then back to the clock and west to the cottage. In the hallway there is a spear, then south to the kitchen to find some catfood in the cupboard. Back to the hallway and up the stairs to the landing. West of the landing is the bedroom and the JEWELLERY CASE, north of the landing is the storeroom and a rope, rucksack and pile of rubbish. The rucksack should be worn to enable you to carry more and the rubbish can be searched to find an object you will need to open the catfood. South and down to the hallway and then west to the living room where you will find a looking glass and the METRONOME, west again takes you to the patio.

From the patio you can go NW to the flower garden, SW to the shed or west to the fence at the end of the garden. Get the valerian from the flower garden first and then to the fence and up, to get to the other side. North from here takes you to the narrow part of the stream where you should use the rope to join the planks and then drop them to enable you to cross the stream. Here you will find the narcissus, give him the looking glass and he will give you the LODESTONE. Back across the stream and south to the fence, up, east and SW to the shed. There is a bunch of keys under the shed, use the LODESTONE to get them. Inside the shed you will find an axe and a shovel. Back to the fence, up and south to the willow tree by the polluted stream. Wave the valerian and then use the axe. Your reward will be the DIAMOND TEARDROP, then go back to the shed.

From the shed go east to the gate and east again to the paved path and a compost heap, search this twice to find a mushroom ring and a silver coin. Eat the mushroom ring, then try and get the silver coin, when you do this a fairy appears and grabs back the coin, demanding you bring her the tooth (time zone 2). We will get to that later. Go east from here to the carport and search the car to find a can of petrol, then go down to return to the clock.

TIME ZONE 2.

Take with you:- MATCHES, CANDLE, SPEAR, PICK, CATFOOD, TIN OPENER and the CAN OF PETROL.

At the start of time zone 2, the clock is at the base of a ice sheet. North 3 times will take you to the edge of the ice where you will find some wood and a mammoth. Pour the petrol by the wood, light a match and then light the wood.

When you light the wood the fire scares off the mammoth leaving behind a fur coat and the IVORY TUSK. Wear the fur coat and go east to the rocky track and then head up the snow mountain to find a bone at the armchair hollow and up again to the ledge outside the cave where you will find the tiger. Open the catfood with the tin opener, give the catfood to the tiger and then pull the tigers tooth.

Light the candle and go in to the cave to a chamber, from here its south to the freezing cave where you will find an icicle. You must make some sort of noise to free the icicle and then back to the chamber and east to the silver fox lair where you should give the bone to the fox to enable you to pass him. Then east to the silver cave and the wall of ice, break the wall to make a hole and go south to the silver vault where you find a CANDELABRA. Back to the silver fox lair and south to the frozen forest where you should drop the icicle and the go NE to the throne room. Here you will find a cube of ice which you must break, the queen will then give you a sword. Go south to the pool of icy water and go in, then west, down and down again to return to the clock.

When you are back at the clock you will then need to return to time zone 1, taking the the tooth with you. Go straight to the compost and give the tooth to the fairy, she will make you an offer, refuse the first offer but accept the second. She gives you a firefly, which is a source of light, there is no need to carry the candle and matches any further.

TIME ZONE 3.

Take with you:- only the FIREFLY.

From where you start go east twice to the Tyranosaurus Rex by the pool of muddy water, going west will lead the t.rex to the prairie where you find an allosaurus, the two of them will take care of each other. Then west again to the deep pit and the brontosaurus, get the leaves and mirror, wave the leaves and then wave the mirror. Go back to the small hill where you left the clock and head up the path and into the cave, when you see the cavemen drop the mirror and go north 3 times picking up the DINOSAUR EGG, (store room) STONE POT (handicraft room) and pebbles along the way, you should now be in the shell room. SE to the painting studio, north to the trophy cave to find a skeleton, throw the pebbles at the skeleton then east to find the ONYX FIGURINE, west twice to the armoury for a stone club then back east to the trophy cave. From here its SE to the wooden bridge and a caveman and cavewoman, throw the club to save the cavewoman and in return she gives you a gold nugget (needed for time zone 5). Then its on to the invention room where a door blocks your way,, but an inscription on the wall will help you to know what to SAY to open the door. In the invention alcove another door poses the same problem and another inscription will also help you here. IN once again and the invention cupboard where you find the WHEEL. Then its down to get back to the clock.

LORDS OF TIME will continue next month, in the meantime if you have any problems in an adventure just drop me a line and I will try to be of assistance. Till next time:

Bye for now, ALAN.

SWAPPING WORDS AND PEOPLE

Hello once again. Here I am with some more "BITS" of information from around Australia and the Globe.

FROM ENGLAND:

You could be forgiven, living in Australia, for thinking MSX/SPECTRAVIDEO computers were dead, what with non-existant backup and very little Hardware/Software available from retailers. However, in the U.K. and Europe MSX is still going strong. In fact, MSX2 is starting to make it's presence felt over there. To give you some idea what's happening over in the 'OLD COUNTRY', a new magazine has just started up, titled "MSX MAGAZINE". Some of the contents are said to include Hints, Pokes, Maps and even a Pen Pal page. Also here are a few New titles just released in the U.K.

THE GAMES ---------- WINTER EDITION (SPORTS)
THE FLINTSTONES --------- (YABBA DABBA DOO)
SHACKLED ---------- ADVENTURE
VENOM STRIKES BACK ------ ADVENTURE

FROM JAPAN:

MSX is still the No.1 home computer in Japan, over One Million (1,000,000) of a game called "Dragon Quest" were sold to MSX users. I guess who could infer from that, that the Japanese like games about Dragons.!!!
Did you know that in Japan, Household electrical items such as Phones and VCR's are fitted with MSX chips internally, to allow preprogramming and remote use by telephone. They really take MSX seriously.

FROM VICTORIA:

Specticale of Victoria, otherwise known as the Melbourne SVI/MSX users group, are a thriving club. They do a tremendous lot of programming work on Disc routines and some assembling. Their aim is to make computer use a lot easier, and fun.

FROM CANADA:

Remember last time I mentioned the Canadian group "SMUG" which issues a Bi-lingual newsletter, well, the Editor, Francis Fontaine writes the English text and also types in the translations French. She also writes articles and reviews for the magazine. Sounds like a lot of hard, tedious work. Just as well we only have one official language.....Aussie.

FROM BRISBANE:

Just heard that there is a shop in Brisbane selling 48K Pioneer MSX Computers very cheap. It shows you there are still some MSX bargains left.

FROM GERMANY:

The good news to report is that a new Bi-lingual MSX magazine has just started up, it is called "THE MSX MAG" and is written in conjunction with the MOS users club, it is written in both German and English. From the first issue it looks like this is going to be a bumper issue full of MSX hints and tips etc..

I am always on the lookout for any interesting MSX news and information, If anyone wishes to contribute please write to me C/- the following address and I will include it in this column.
NOEL BERGMAN, 16 SURREY ROAD, BELLBIRD PARK, QLD 4300 AUSTRALIA.

Well that wraps up S.W.A.P. for this issue.. Bye for now NOEL

RACE

Here is a small Horse Racing game for MSX owners but with a few minor modifications it will run on the SV318 & SV328 computers.

```
BE  10 CLS:KEY OFF:WIDTH 39:RDRST=R
    ND(-TIME)
GI  20 PRINT"  ------- FORUM RACE
    TRACK ---------"
CP  30 PRINT:PRINT:PRINT"You start
    off on race day with $500.00 in
     your pocket curtesy of
          COMPUTER FORUM"
MB  40 PRINT:PRINT"There are 8 hors
    es and 10 races, all   you have
     to do is pick the correct
    horse."
CI  50 PRINT:PRINT"Sounds easy to m
    e.. GOOD LUCK."
CN  60 '
EG  70 PRINT:INPUT"  Press [RETURN]
     to continue:";ZZ
FC  80 CLS
DB  90 INPUT"How many players:";P
CB  100 FOR X=1 TO P
BL  110 PRINT:PRINT"Name of player
    :";X;:INPUT A$(X)
DG  120 NEXT X
BM  130 CLS
DO  140 LR=INT(RND(1)*20)+1
CE  150 IF LR<10 THEN 140
BB  160 PRINT"Length of Race:";LR;"
     furlongs":PRINT
DH  170 PRINT"RACE - 1st - 2nd - 3r
    d - FURLONGS"
CG  180 FOR X=1 TO 10
EF  190 PRINT X;TAB(7)WI(X);"   ";S
    E(X);"   ";TH(X);"     ";LR(X)
DD  200 NEXTX
CB  210 FOR X=1 TO P
DB  220 IF BO(X)=0 AND MO(X)<1 THEN
     MO(X)=500:BO(X)=1
AM  230 LOCATE 0,14:PRINT SPC(200)
EK  240 LOCATE 0,14:PRINT X;" ";A$(
    X);" you have $";MO(X)
EA  250 LOCATE 0,16:PRINT"Pick a ho
    rse from 1 to 8:";:INPUTPH(X)
DJ  260 IF PH(X)<1 OR PH(X)>8 THEN
    BEEP:LOCATE 27,16:PRINT"    ":
    GOTO 250
BC  270 LOCATE 0,17:INPUT"Bet how m
    uch eg 50      :";B(X)
DJ  280 IF B(X)>MO(X) THEN MO(X)=MO
    (X)-B(X)
EN  290 NEXT X:X=0
BH  300 CLS
AN  310 IF RA<1 THEN RA=1
AG  320 LR(RA)=LR
EN  330 PRINT TAB(25);"Race #:";RA
DC  340 FOR T=1 TO 18:LOCATE LR,T+1
    :PRINT"|":NEXT T:T=0:V=1
CB  350 C=C+1
FD  360 X=X+2:LOCATE 0,X+1:PRINTC
BM  370 IF C=8 THEN 380 ELSE 350
CM  380 C=0:X=0:MH=INT(RND(1)*LR)+1
CN  390 LOCATE 27,8:PRINT"1st:":LOC
    ATE 27,10:PRINT"2nd:":LOCATE 27
    ,12:PRINT"3rd:"
AB  400 IF V=1 THEN V=0:GOSUB 770
CK  410 IF MH=1 THEN H1=H1+1:HN=H1:
    D=3:GOSUB 500
CH  420 IF MH=2 THEN H2=H2+1:HN=H2:
    D=5:GOSUB 500
CE  430 IF MH=3 THEN H3=H3+1:HN=H3:
    D=7:GOSUB 500
CB  440 IF MH=4 THEN H4=H4+1:HN=H4:
    D=9:GOSUB 500
BF  450 IF MH=5 THEN H5=H5+1:HN=H5:
    D=11:GOSUB 500
BE  460 IF MH=6 THEN H6=H6+1:HN=H6:
    D=13:GOSUB 500
BD  470 IF MH=7 THEN H7=H7+1:HN=H7:
    D=15:GOSUB 500
BC  480 IF MH=8 THEN H8=H8+1:HN=H8:
    D=17:GOSUB 500
AD  490 GOTO 380
GB  500 LOCATE HN-1,D:PRINT " ";MH;
DH  510 IF HN=LR THEN W=W+1 ELSE 55
    0
CI  520 IF W=1 THEN LOCATE 31,8:PRI
    NTMH:WI=MH:WI(RA)=MH
DF  530 IF W=2 THEN LOCATE 31,10:PR
    INTMH:SE=MH:SE(RA)=MH
DB  540 IF W=3 THEN LOCATE 31,12:PR
    INTMH:TH=MH:TH(RA)=MH
AF  550 IF W=3 THEN 560 ELSE RETURN
AO  560 LOCATE 0,22:PRINT:INPUT"Pre
    ss [RETURN] to continue:";ZZ:W=
    0:H1=0:H2=0:H3=0:H4=0:H5=0:H6=0
    :H7=0:H8=0
CE  570 FOR X=1 TO P
JA  580 IF PH(X)=WI THEN MO(X)=MO(X
    )+B(X)*4
IO  590 IF PH(X)=SE THEN MO(X)=MO(X
    )+B(X)*3
JJ  600 IF PH(X)=TH THEN MO(X)=MO(X
    )+B(X)*2
AL  610 IF PH(X)<>WI OR PH(X)<>SE O
    R PH(X)<>TH THEN MO(X)=MO(X)-B(
    X)
DB  620 NEXT X
BK  630 IF RA=2 THEN 640 ELSE RA=RA
    +1:GOTO 130
AL  640 '
```

```
CG  650 CLS:PRINT"Race day is over.
     Here's the results"
CC  660 FOR X=1 TO P
ED  670 PRINT:PRINT A$(X);" has $";
    MO(X);".00"
EM  680 IF MO(X)<0 THEN AO(X)=ABS(M
    O(X)):G$(X)=A$(X)+" Owes COMPUT
    ER FORUM $"+STR$(AO(X))+".00":O
    M=1
DI  690 NEXT X
CP  700 IF OM=1 THEN 710 ELSE END
CK  710 PRINT:PRINT
BN  720 FOR X=1 TO P
AD  730 PRINT G$(X)
DC  740 NEXTX
BO  750 PRINT:PRINT"A CHEQUE WOULD
    BE FINE.."
AK  760 END
BK  770 LOCATE 0,19:PRINT"PLAYER: 1
     - HORSE:";PH(1)
CC  780 LOCATE 0,20:PRINT"        2
     - HORSE:";PH(2)
CC  790 LOCATE 0,21:PRINT"        3
     - HORSE:";PH(3)
BA  800 LOCATE 23,19:PRINT"4 - HORS
    E:";PH(4)
AH  810 LOCATE 23,20:PRINT"5 - HORS
    E:";PH(5)
AJ  820 LOCATE 23,21:PRINT"6 - HORS
    E:";PH(6)
CF  830 RETURN

          THE END
```

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MSX BASIC TUTORIAL PART 7

Before I start off explaining the RACE game let me explain why I did not write up the word-processor program for this issue.

It is a very simple reason. I started making the program with what I thought was the barest of essentials required for such a program, and then it just got bigger and bigger until it was far to large a program to put into this column.
So then I came up with a HORSE RACE game which is quite good.

Again I will highlight the commands I think you should know on certain lines.

```
90 INPUT"How many players:";P
```
This line sets up the number of players in the game, I have not put it in a limit of players but you should. (Hint P>6 THEN)

```
100 FOR X=1 TO P
```
A counter to the number of (P)layers

```
110 Name of Player";(X);:INPUT A$(X)
```
This line sets up the name of player (X) or 1 the first time through the loop and stores it under A$(X)

```
140 LR=INT(RND(1)*20)+1
```
LR is the Length of the Race, and is picked out from a RaNDom number.

```
150 IF LR<10 THEN 140
```
This line specifies that the length of the race must be greater than 9 furlongs.

```
190 PRINT X;TAB(7)WI(X);"   ";SE(X);"   ";TH(X);"  ";LR(X)
```
This line sets up the winnings board, what horse came First, Second, Third and how far they ran.

```
220 IF BO(X)=0 AND MO(X)<1 THEN MO(X)=500:BO(X)=1
```
This line checks to see if the player has any money and if not gives him 500 and then sets a counter so it does not give him any more:BO(X)=1

```
230 LOCATE 0,14:PRINT SPC(200)
```
This line sets the cursor at ZERO columns across by 14 rows down and then prints 200 blank spaces which clears the bottom part of the screen.

```
250 PRINT"Pick a horse.etc ";:INPUT PH(X)
```
This line does exactly what it says, pick a horse and then it stores your selection under PH(X).

```
270 INPUT"Bet how much eg 50 :";B(X)
```
As the line suggests, how much do you wish to bet and it also stores your answer under a variable, B(X).

```
310 IF RA<1 THEN RA=1
```
If the RAce is less than(<) RAce 1 then make it RAce 1

320 LR(RA)=LR
This result of LR(RA) is used in line 190 to tell you how long that race was. What it actually says is- If for instance RAce=1 then LR(Length of race)(RA)= whatever number is picked out on line 140

340 FOR T=1 TO 18:LOCATE LR,T+1:PRINT" ":NEXTT:T=0:V=1
This line draws the finish line, LR(across) by T(down)

350-370 Printup the horse numbers.

380 C=0:X=0:MH=INT(RND(1)*LR)+1
This line picks the RaNDom horse to run.
If you wish you can change the LR in this line to 8 and that will speed the horses up. I made it LR so the race is a bit slower.

390 LOCATE 27,8:PRINT"1st" etc etc..
This line prints up 1st, 2nd and 3rd on the screen.

410-490 IF MH= etc..
If in line 380 MH=1 then horse 1 will move, if MH=2 then horse 2 will move and so on.
Say MH=5. Line 450 would be saying--
H5=H5+1(Horse no5):HN(Horse No)=H5:D=11(Down):GOSUB500

500 LOCATE HN-1,D:PRINT " ";MH
This line prints the horse number across the screen. In other words - LOCATE HN(Horse No)-1 the minus 1 is used to start the horse from the far left.,D(Down) the D or Down is set up by whatever horse is picked, in our example above HN=1 and D=11.

510 IF HN=LR THEN W=W+1 ELSE 550
This line checks to see if any horses have crossed the line. IF HN(Horse No)=LR(Length of Race) then W=W+1

520-540 IF W= etc ...
If W is GREATER than 1 then these lines come into affect. Line 520 sets up the WINNER and stores the number under WI(RA)

570-620 FOR X=1 TO P etc...
These lines are set up to work out your winnings or your loss.
580 IF PH(X) (Horse player one picked)= WI (Winner) THEN MO(X) (Money of player 1)=MO(X)+B(X) (Bet of player 1)*3 (Times 3).

630 IF RA=2 THEN 640 ELSE RA=RA+1:GOTO 130
This line sets up the number of RAces.

650-750 Prints up all the race results.

There seems to be a bug in the program somewhere, I'm sure you will find it.

Well until next time bye....

Beyond Basic (Part 2)

by Tony Cruise

In this section I will explain how you can attach extra commands to ROM routines.

In MSX computers there are many Hook Jumps provided so that machine code programmers can attach thier own routines to most of the basic ROM routines. Each Hook consists of five memory locations in RAM. When the computer is first switched on or reset, each of the locations contains the value 201. This is the decimal value for a RETURN statement in machine code. Each of the Hooks are jumped to at the start of each of the corresponding ROM routines, so you can insert or eliminate entirely each routine.

To use a Hook, all you have to do is insert your own commands into the five memory locations. This is usually a JUMP instruction to your machine code routine.

Here is an example program that uses the Hook **HTIMI**. This Hook is jumped to every 1/50th of a second.

Machine Code Program

```
D000            1 ORIGIN ORG 0D000H
                2 ;
                3 ;  PROGRAM TO UPDATE AND DISPLAY A CLOCK
                4 ;  ON SCREEN 0
                5 ;
                6 ;  LABEL SETTINGS
                7 HTIMI  EQU 0FD9FH ; Hook Jump - Timer
                8 SCRMOD EQU 0FCAFH ; RAM Storage - Screen Mode
                9 SETWRT EQU  0053H ; ROM Routine - Set Video Write
               10 ;
D000 3EC3       11 START: LD A,0C3H      ;Value for JUMP instruction
D002 329FFD     12        LD (HTIMI),A   ; Load into Hook
D005 210CD0     13        LD HL,CUE      ; Address to jump to
D008 22A0FD     14        LD (HTIMI+1),A ; Load into Hook
D00B C9         15        RET
               16 ;
D00C CD14D0     17 CUE:   CALL CLOCK     ; Call CLOCK routine
D00F F7         18        RST 030H       ; Disk delay routine
D010 879C77     19        DEFB 087H,09CH,077H ;-remove if you do not
D013 C9         20                            ; have a disk drive
               21 ;
D014 161D0      22 CLOCK: LD HL,STORE    ; Decrease counter
D017 7E         23        LD A,(HL)      ;
D018 3D         24        DEC A          ;
D019 77         25        LD (HL),A      ;
D01A C0         26        RET NZ         ; Return if not ZERO
D01B 3E33       27        LD A,51        ; Set new counter
D01D 77         28        LD (HL),A      ;
D01E 2162D0     29        LD HL,CLKSTR   ; Update - seconds
D021 0603       30        LD B,3         ;        - minutes
D023 7E         31 CL1:   LD A,(HL)      ;        - hours
```

****

```
D024 3C      32          INC A           ;
D025 27      33          DAA             ; Decimal addition
D026 FE60    34          CP 060H         ;
D028 3003    35          JR NC,CL2       ;
D02A 77      36          LD (HL),A       ;
D02B 1805    37          JR PRTCLK       ;
D02D 3600    38 CL2:     LD (HL),0       ;
D02F 23      39          INC HL          ;
D030 10F1    40          DJNZ CL1        ;
D032 3AAFFC  41 PRTCLK:  LD A,(SCRMOD)   ; Get screen mode
D035 3D      42          DEC A           ;
D036 D0      43          RET NC          ; Check for screen 0
D037 211F00  44          LD HL,31        ;
D03A CD5300  45          CALL SETWRT     ; Set screen address
D03D 2164D0  46          LD HL,CLKSTR+2  ; Set pointer to clock
D040 0603    47          LD B,3          ; Set counter
D042 7E      48 P1:      LD A,(HL)       ;
D043 27      49          DAA             ;
D044 F5      50          PUSH AF         ; Save value
D045 CB3F    51          SRL A           ; Get right digit
D047 CB3F    52          SRL A           ;
D049 CB3F    53          SRL A           ;
D04B CB3F    54          SRL A           ;
D04D C630    55          ADD A,48        ; Change to ASCII
D04F D398    56          OUT (098H),A    ; Send to screen
D051 F1      57          POP AF          ; Return value
D052 E60F    58          AND 1111B       ; Get left digit
D054 C630    59          ADD A,48        ; Change to ASCII
D056 D398    60          OUT (098H),A    ; Send to screen
D058 3E3A    61          LD A,58         ; Value for ':'
D05A 2B      62          DEC HL          ; Update pointer
D05B 05      63          DEC B           ; Update counter
D05C C8      64          RET Z           ; Return if finished
D05D D398    65          OUT (098H),A    ; Send ':' to screen
D05F 18E1    66          JR P1           ; Go again
             67 ;
             68 ; DATA STORAGE
D061 33      69 STORE:DEFB 51
D062 000000  70 CLKSTR:DEFB 0,0,0
D065         71 END:
```

Basic Loader

```
10 CLS:CLEAR 200,&HCFFF:DEFINTA-Z:A=&HD000
20 READ A$:IF A$<>"@" THEN POKE A,VAL("&H"+A$):A=A+1:GOTO 20
30 PRINT"  INSERT TAPE/DISK TO SAVE PROGRAM"
40 PRINT"       AND PRESS ANY KEY"
50 A$=INPUT$(1):PRINT:PRINT"  SAVING ....."
60 BSAVE"CLOCK",&HD000,A-1
100 DATA 3E,C3,32,9F,FD,21,0C,D0,21,0C,D0,22,A0,FD,C9,CD
110 DATA 14,D0,F7,87,9C,77,C9,21,61,D0,7E,3D,77,C0,3E,33
120 DATA 77,21,62,D0,06,03,7E,3C,27,FE,60,30,03,77,18,05
130 DATA 36,00,23,10,F1,3A,AF,FC,3D,D0,21,1F,00,CD,53,00
140 DATA 21,64,D0,06,03,7E,27,F5,CB,3F,CB,3F,CB,3F,CB,3F
```

****

```
150 DATA C6,30,D3,98,F1,E6,0F,C6,30,D3,98,3E,3A,2B,05,C8
160 DATA D3,98,18,E1,33,00,00,00,@
```

If you do not have a disk drive, change line 110 to -:

```
110 DATA 14,D0,00,00,00,00,C9,21,61,D0,7E,3D,77,C0,3E,33
```

This program displays a clock in the top right hand corner of the screen, but only when the computer is in screen mode zero. The first part of the program fills the Hook HTIMI with a jump command pointing to a routine called CUE. This routine calls all the things that are using this Hook. In this case, just the two things are being called, the clock routine, and the routine to turn the disk drive motor off after a certain period of time. If you are not using a disk drive, set the four memory locations to zero. In later articles, I will detail more routines that use this Hook.

The clock routine starts by decreasing the first counter, which was initially set to 51. This counter is to measure the passing of one second (The Hook is called every 1/50th of a second). Once a second has passed, the clock counters for the hour, minutes and seconds are updated. Then, if the screen mode is set to zero, the clock will be printed out.

In the next section, I will cover another of the Hook jumps called **HGONE**, and show how new statements can be added to your Basic programs.

OOPS !! - Last issue, due to a type setting error, there was a mistake in the first line of the Basic loader program. The line should have read as follows -:

```
10 CLS:CLEAR 200,&HCFFF:DEFINTA-Z:A=&HD000
```

****

SPECTRAVIDEO SV-328 OWNERS .... are you interested in a Five Meg-a-byte Hard disk for your expander? Possibly expandable to Ten Megs, under CP/M. If so, please drop me a line.

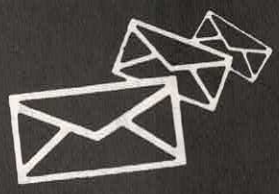
I have spoken to a former SV-328 owner who may be able to build a Hard disk and Controller, if we have enough numbers to make it economically feasible. So please put pen to paper, or fingers to the keyboard, and let me know if you are interested at all.

If anyone has any knowledge of any other things that could be made for the SVI or MSX computers, (eg Real Time Clock card, Video Digitizer, Modem, Speach Module etc.) could you also let me know so as I can pass on the information to our other readers

The address to write to is -:

ADD ON BITS AND PIECES
C/- MAX COWIN,
PO BOX 87,
MtHAWTHORN, W.A. 6016.

MAILBOX



A letter from Mark Sands of Tasmania.

Well, to start again, I have already typed this letter once tonight, but I got a "DISK FULL" message from my Wordpro when I went to save it, and as a result, I lost it. I use the Write program from the Thinkman package. It's not bad, but I feel it needs two disk drives to be in operation.

First I would like to congratulate you on helping to keep MSX and Spectravideo alive in Australia, keep up the good work. I have owned a Sony Hit Bit for three years and I still think we have the best basic available on a 64k computer.

I read with interest the article on MSX Technical books and, if you get more than fifty people interested, then I would like two of MSX-TDB, one of MSX-BIOS and one of MSX-VDP.

If you would not mind doing so, I would like my name, address and phone number printed in the magazine with a request that anyone in Tasmania interested in starting a user group, contact me AFTER NORMAL BUSINESS hours. This would be much appreciated.
I also own a Interlink modem, but I have neither hooked it up nor am I sure of how to use it. Any help with this would be appreciated.

Now for some input for the magazine in the way of games addresses. Most are made up of more than one part and I will write the game name followed by the program names and addresses. Most won't work on disk because of thier location in memory.

NAME	PROGRAM	START	END	RUN ADDRESS
GUNFRIGHT	GUN	&HD100	&HEE45	&HD100
	A	&H9000	&HD000	&H9000
	B	&H9000	&HD000	&H9000
ALIEN 8	ALIEN	&HD100	&HEAF5	&HD100
	A	&H9000	&HD000	&H9000
	B	&H9000	&HD000	&H9000
KNIGHTLORE	KNIGHT	&HD800	&HE94F	&HD800
	A	&H9000	&HD000	&H9000
	B	&H9000	&HD000	&H9000
HUNCHBACK	HUNCH	&H9000	&HD000	&H9000
HUSTLER	HUS	&HC000	&HEFFF	&HC3B9

****

Thank you for your letter, Mark. And as you requested here is your name and address in the magazine:-

MARK SANDS
48 HERBERT St.,
INVERMAY,
TASMANIA 7248.
PH (003) 340552

If anyone in Tasmania is interested in starting a user group, please give Mark a ring, and remember to call AFTER business hours.
Thank you for the games address's, I'm sure our readers can use them to back-up thier own games.

As to the Interlink Modem, I had a small try with this modem and found it quite easy to use. First off, you need a telephone line. The modem plugs into your cartridge port and also connects to your phone line. I put a phone socket in "my room" so that I would not have to move my computer and screen into the family room (thats where the phone is).
Once the computer, modem and phone line are all connected, you then turn your computer on, the next step is to make sure the modem is set up correctly for the job you wish it to do.(The instruction manual has all the information you need to do this on page 11.)

The Modem has built-in firmware to handle Viatel and normal communications using 300 baud Full Duplex and 1200/75 split baud rate.
To access Viatel (see page 14 of the manual), type CALL VIDEOTEX and set the top silver switch to phone, and the bottom silver switch to originate. Dial the Viatel number and when the carrier signal is heard (a high pitched whistle), set the top switch to modem. At this point, the introductory Viatel screen is displayed and you are asked for a password. If you do not have a Viatel account you will not get passed this screen. From this point on, you follow the prompts and go to the section that you are after.

To use the other firmware program COMTERM (page 19, para 8.6), you need to find the telephone number of a Bulletin Board System (BBS) or a friend with another modem. If you call a BBS, then it is the same as for Viatel, except you use COMTERM and the set up is for 300/300 board. If you are calling a friend, one of you sets up for Originate and the other for Answer or Terminate. Then both of you switch your modems from phone to modem and you are away. (Of course, you have CALLed COMTERM already)

In a previous issue (V2 No3 P3) Tony (our Secretary) asked for anyone with modems to contact him with the intention of starting up a MODEM CLUB. As we have not had any response to this, it was believed that there is no interest in using the computer, via the modem and phone line, to communicate with others. We hope that this is changing, as the use of modems can help speed up the transfer of data and, if using the correct software, files of different types and sizes.

A query from Phillip Christie, he writes-

Does anyone know if the JAN/FEB 1987 copy of "MSX COMPUTING" was the last edition published. There was a couple of errors in the program "TITAN LAND" which was published in that edition, and if there was another edition, did they explain the errors in this program??

REPLY

As far as we know the last issue of "MSX COMPUTING" was the JAN/FEB issue due to some rather strict tax laws.
In regards to the program TITAN LAND, I think every man and his dog has tried to fix up this program with no success.
If by any chance someone has debugged it and has a working copy please let us know, and perhaps we could publish the incorrect lines or even send out copies to all concerned.

CARTRIDGES

As you know we used to sell CARTRIDGES through our magazine, at the moment our source has dried up, we used to buy the cartridges through SONY (Sydney) at wholesale prices, add $5.00 for Postage and packaging and sell them to you.
Because of the slow turnover 14 Cartridges in 7 months, SONY decided to sell all of it's software in a fire sale at a reduced price. We thought it was to high priced anyway.

THE MYSTERY OF LETHE CASTLE

I was hoping for a better response to ALAN's adventure game "THE MYSTERY OF LETHE CASTLE", there must be more adventurers out there who would like this game??

This game is on the same sort of level as most Adventure games except for one minor detail it is half the price.
I think that $15.00 is extremely good value for the effort and time placed into LETHE CASTLE, so what about supporting ALAN & this magazine and buy the game.

BACK ISSUES

We are trying to get up some money to have a competition on a large scale but we need a few hundred dollars first, this is where BACK ISSUES come into effect.

We have all back issues except for Vol 1:No 7 & Vol 1:No 8 and we are selling these for $2.00 per copy.

If you have been thinking of buying back issues, then now is the time.
Please help us to help you.

Dear SVI 318/328 users,

Have you ever been annoyed by the fact that some commands terminate program execution? eg CLOAD,MERGE, DELETE, LIST etc. Well LIST has no excuse, but the others modify the program length, which upsets the variable table, but this is not necessarily a problem, ie if you have a piece of code that must be executed once only, regardless of how many times the program is run, you have it DELETE itself, but that means prompting the user to type 'RUN' again (Very user UNfriendly). Well the answer is in the new command I've invented - **"AUTOKEY"**. AUTOKEY automatically puts key strokes into the hopper, just as if the user typed them. Then, when the program stops, the key strokes are executed. AUTOKEY takes the form - "AUTOKEY(string expression)" - as in the example below.

```
5 GOTO 60
10 PRINT"This is the part of the program that"
20 PRINT"can be run as many times as you want,"
30 PRINT"but only after lines 60 & 70 have"
40 PRINT"been executed first."
50 END
60 AUTOKEY"DELETE 5 "+CHR$(13)+"RUN"+CHR$(13)
70 MERGE" CAS:DATA"
```

During the first execution, line 5 directs to line 60 where AUTOKEY puts "DELETE 5 (ENTER) RUN (ENTER)" into the hopper, then MERGE is executed. After which, execution stops, line 5 is then DELETED, and the program is RUN starting at line 10.

I have included both the assembly listing and a short loader in BASIC which can be MERGED with your programs, and is self DELETEING. You may locate the machine code any where you like by changing the CLEAR statement and the "AD" assignment.

I hope many of you find this as usefull as I have.

Martin Everitt, Eagleby, QLD.

BASIC PROGRAM

```
1000 CLEAR 500,&HEFFF:AD=&HF000:RESTORE1100:AC=AD
1005 READ A$:IFA$<>"END" THEN POKE AC,VAL("&H"+A$):AC=AC+1:
     GOTO1005
1006 PRINT"NEXT FREE ADDRESS =";HEX$(AC)
1010 HK(0)=&HC3:HK(2)=VAL("&H"+LEFT$(HEX$(AD),2)):HK(1)=AD AND
     &HFF:POKE AD+&H32,(AD+&H39) AND &HFF:POKE AD+&H33,VAL("&H"
     +LEFT$(HEX$(AD+&H39),2))
1020 FOR=2TOOSTEP-1:POKEAD+&HC+X,PEEK(&HFF57+X):POKE&HFF57+X,
     HK(X):NEXTX
1030 PRINT"READY":AUTOKEY"RUN"+CHR$(13):DELETE1000-1100
1040 PRINT"FAULT"
1100 DATA FE,A9,20,8,23,7E,FE,C7,28,5,2B,7E,C9,0,0,33,33,23,CD,
     CA,14,E5,3A,93,F7,FE,3,20,14,2A,25,F9,46,23,5E,23,56,C5,1A,
     CD,A,40,C1,13,10,F7,E1,7E,C9,21,39,DO,CD,EA,11,18,F5,31,33,
     3A,END
1110 PRINT"IT WORKS":DELETE1110
```

****

MACHINE CODE

```
F000 FEA9     CP   0A9
F002 2008     JR   NZ,0F00C
F004 23       INC  HL
F005 7E       LD   A,(HL)
F006 FEC7     CP   0C7
F008 2805     JR   Z,0F00F
F00A 2B       DEC  HL
F00B 7E       LD   A,(HL)
F00C C9       RET
F00D 00       NOP
F00E 00       NOP
F00F 33       INC  SP
F010 33       INC  SP
F011 23       INC  HL
F012 CDCA14   CALL 14CA
F015 E5       PUSH HL
F016 3A93F7   LD   A,(F793)
F019 FE03     CP   03
F01B 2014     JR   NZ,0F031
F01D 2A25F9   LD   HL,(F925)
F020 46       LD   B,(HL)
F021 23       INC  HL
F022 5E       LD   E,(HL)
F023 23       INC  HL
F024 56       LD   D,(HL)
F025 C5       PUSH BC
F026 1A       LD   A,(DE)
F027 CD0A40   CALL 400A
F02A C1       POP  BC
F02B 13       INC  DE
F02C 10F7     DJNZ 0F025
F02E E1       POP  HL
F02F 7E       LD   A,(HL)
F030 C9       RET
F031 2139F0   LD   HL,F039
F034 CDEA11   CALL 11EA
F037 18F5     JR   0F02E
F039 31333A   LD   SP,3A33
```

Our thanks go to Martin for sharing his new command with us. I'm glad to see someone overcoming a problem like this, and then sharing his solution around. Keep up the good work Martin.

VERY DISAPPOINTED

SOFT-ADDS

We have had a few enquiries about SOFT-ADDS, "Where has it gone" we decided to scrap it because of the lack of response, we sold about 20 copies of the games for sale, and that was from when we started it early last year.
We will now send out cheques to those people involved.

MSX PACKS

Are these packs to face the same fate as SOFT-ADDS???
As you know we do not have any paid advertising, so all of our money comes from either Subscriptions or Software. At the moment we are running on subscription money which is not a good position to be in with rising costs.

Please.. again I ask, help this magazine survive and buy our software.

" DESIGN A SPRITE "

By FRANK KEAN

We can have up to 32 sprites on the screen at the same time, provided each sprite is on a different plane. The sprites can be all the same, all different or some the same and some different. Each sprite is made of 64 squares, each square being one pixel on an 8 pixel x 8 pixel square grid. To design a sprite we draw a grid and fill in the squares to give us the shape we want, (fig.1). For the computer to fill a square in we have to tell it that the square is 1. To have a square blank we have to tell the computer that the square is 0, (fig.2). The sprite used here is one I used for a rough spacehip and it is also one of the block graphic characters, the "spade", which we can get by pressing "RIGHT GRPH" & "V" (SV Units) at the same time.

FIG.1

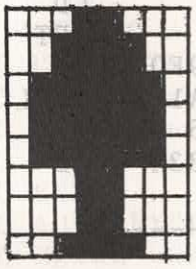


FIG.2

```
00011000
00111100
01111110
01111110
01111110
00011000
00011000
00111100
```

Using the system in the S.V. Manual we can write the following program. (List 1), then <run> it to see our sprite.

```
5   '     LIST 1
10  COLOR 1,15,7:SCREEN 1 (Screen 2 for MSX)
20  FOR T=1 TO 8:READ DA$
30  S1$=S1$+CHR$(VAL("&B"+DA$)):NEXT
40  SPRITE$(1)=S1$
50  PUT SPRITE 0,(128,96),1,1
60  GOTO 60
100 REM DATA FOR SPRITE
110 DATA 00011000
120 DATA 00111100
130 DATA 01111110
140 DATA 01111110
150 DATA 01111110
160 DATA 00011000
170 DATA 00011000
180 DATA 00111100
```

Line 10 sets the colour to 1 = black for text, 15 = white for background, 7 = cyan for border: Screen 1 for high resolution.

Line 20 reads the data, using a "FOR : NEXT" loop of 8 times to read the 8 Data statements (lines 110 to 180). In this case each data statement is one line.

Line 30 builds the sprite. DA$ is the information in each data statement. (In this case each line of data). "&B" tells the computer that what follows (DA$) is a binary number. VAL tells the computer to find the VALue of what follows. CHR$() tells the computer to find the CHaRacter STRING of what follows. S1$ is the name of this sprite design. NEXT at the end of the line completes the loop. In this loop the sprite is built by adding each data statement together, so that

```
S1 = CHR$(VAL("&B" + 00011000))
   + CHR$(VAL("&B" + 00111100))
   + CHR$(VAL("&B" + 01111110))
   + CHR$(VAL("&B" + 01111110)) & etc. for the 8
                                  statements.
```

Line 40 makes sprite$(1) (ie. sprite No1) = to the design called S1$. (In line 50 sprite No1 is designed by the second 1. Do not confuse this sprite 0, which indicates the plane number).

Line 50 places the sprite in position on the screen, in this Case it is in the centre of the screen. Put sprite [in plane] 0, [at] (128 [across the screen], 96 [down the screen]),[colour it black] 1, [use sprit No] [ie. design S1$].

Line 60 holds computer in that position.

Line 110 - 180 are data statements from the 8 x 8 square grid.

In this program line 40 could be left out and line 30 would be SPRITE$(1) = SPRITE$(1)+CHR$(VAL("&B"+DA$)):NEXT
If we are only using one sprite this can save a line, however if we are using more than one sprite of the same design it is easier to call up-: SPRITE$(2)=S1$:SPRITE$(3)=S1$ & etc than SPRITE$(2)=SPRITE$(1):SPRITE$(3)=SPRITE$(1) & etc

It is possible to put more than one sprite on one plane but not at the same time. Immediately we put another sprite on the same plane the existing sprite will dissappear. This will give the effect of sprites flashing on and off, which might be of some value for special effects but would not normally be used. To see this effect add lines 41-49,51 and 1000, and change line 50 in list 1 and <run>.

```
41 SPRITE$(2)=S1$
42 SPRITE$(3)=S1$
43 SPRITE$(4)=S1$
48 PUT SPRITE 0,(108,96),7,2:GOSUB 1000
49 PUT SPRITE 0,(118,96),9,3:GOSUB 1000
50 PUT SPRITE 0,(128,96),1,1:GOSUB 1000
51 PUT SPRITE 0,(138,96),13,4:GOSUB 1000
1000 FOR T = 1 TO 100: NEXT: RETURN
```

Now delete lines 41-49,51 and 1000, and delete [:gosub 1000] in line 50, then add

```
51 LOCATE 129,126:PRINT"RG+V"'press RIGHT GRAPH
add V together
52 LOCATE 150,96:PRINT "Sprite"
53 LOCATE 150,126:PRINT"Block Graphics"
```

Run this and we will get our sprite, to screen 2 and <RUN>. When we change the screen statement to screen 2 which is the low resolution screen, our sprite will remain the same but anything else on the screen will go "chunkey", ie. low resolution.
We can stay in high resolution and magnify only the sprites, while all else remains the same size. This is done by changing line 10 to screen 1,1. If we <RUN> the program now the "spade" and printing will stay normal, while the sprite will be magnified and become "chunkey". We can also have a magnified sprite on a low resolution screen. Change line 10 to screen 2,1 and <RUN>. To return to normal change line 10 to screen 1,0 and <RUN>.

Screen 1,0 is normal sprite with high resolution graphics
Screen 1,1 is magnified sprite with high resolution graphics
Screen 2,0 is normal sprite with low resolution graphics
Screen 2,1 is magnified sprite with low resolution graphics

There is no need to state screen 1,0 or screen 2,0 in the first screen statement of a program because our computer will put the ",0" in. However once screen 1,1 or 2,1 is used we must use 1,0 or 2,0 to go back to normal sprite size.

At the begining we said that we can have 32 sprites on different planes at the same time. We have to imagine that we are looking at the television screen from the side, and are able to see 32 thicknesses, one behind the other. Each of these thicknesses is a plane. The planes are numbered from 0 to 31.

The outer plane, that is the plane closest to us if we were in front of the screen looking at it, is No0. The planes are numbered back into the television to the inner most plane, No31. On each of these planes, we are able to place one sprite. If we move a sprite on plane 0 (the outer plane), it will pass over any sprite on planes 1 to 31 (the inner planes), with out being blocked out. If we move a sprite on planes 1 to 31 (the inner planes), it will move under any sprite on plane 0 (the outer plane), and the sprite on planes 1 to 31, will dissappear under the sprite on 0. This means that any sprite on a lower number plane, will move over the top of any sprite on a higher number plane.
Conversely any sprite on a higher number plane will move under any sprite on a lower number plane. To demostrate this delete lines 51-53 and add lines 52-56, 200 and change line 50 then <RUN>. The sprite on plane 1 will move under the sprite on plane 0. Change screen 1,0 to 1,1 and it will be clearer. If we change sprite 0, to sprite 1, in line 50 and change sprite 1, to sprite 0, in lines 52-56 then <RUN>, the sprite on plane 0 will move the sprite on plane 1, as explained above.

```
50 PUT SPRITE 0,(128,96),1,1:GOSUB200
52 PUT SPRITE 1,(108,96),14,1:GOSUB200
53 PUT SPRITE 1,(118,96),14,1:GOSUB200
54 PUT SPRITE 1,(128,96),14,1:GOSUB200
55 PUT SPRITE 1,(138,96),14,1:GOSUB200
56 PUT SPRITE 1,(148,96),14,1:GOSUB200
200 FOR T = 0 TO 500:NEXT:RETURN
```

We will learn more about sprites in the next issue..

USER GROUPS

SYDNEY

We have been advised of a group in Sydney for Spectravideo owners- Please contact Malcolm Samuels on 888 7664 after hours.

QUEENSLAND

BRISBANE SPECTRAVIDEO & MSX USERS GROUP
C/- Mr Noel Bergman (Secretary)
16 Surrey Road.
Bellbird Park Queensland 4300

VICTORIA

MELBOURNE MSX & SPECTRAVIDEO USERS GROUP INC.
C/- Mitch Raitt

We meet on the first Saturday afternoon of each month (except January) at the Nunawading Civic Centre 12.00 - 5.00

---- Phone (03)432 1346 ----

GIPPSLAND MSX USER GROUP
C/- Ross Johnson
---- Phone (051)344 753 ----

WESTERN AUSTRALIA

THE MSX & SPECTRAVIDEO USERS GROUP OF W.A
C/- Max Cowin
PO Box 87 Mt Hawthorn
Western Australia 6016
---- Phone (09)417 3323 ----

Meetings are held on the LAST Tuesday of each month at the Colin st. Hall South Perth (Kensington) at 7.30pm.

NEW ZEALAND

WELLINGTON SPECTRAVIDEO USERS GROUP
PO Box 26-050
Newlands, Wellington
New Zealand 6004

CANADA

SMUG of Montrel
20 - 159th Avenue
N.D. Ile Perrot, Quebec
J7V 5V6 Canada

FINLAND

SVI/MSX Club/Turku
C/- Mika Ristolainen
Lumikonkatu 7H 272 SF-20240
Turka Finland

If any of the above groups wish to enter their meeting times & place or phone number, drop us a line and we will include them in this page.

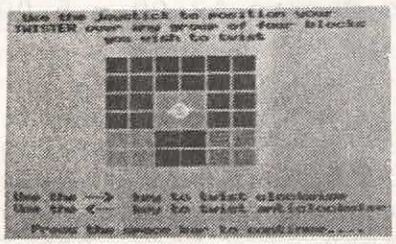
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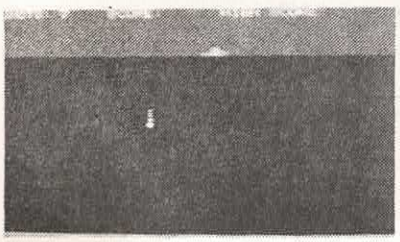
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SVPAK23 - MSXPAK23 -

TWISTER - Like a two dimensional Rubiks cube, this game displays nine different coloured squares each containing four smaller squares. These 36 squares are then jumbled around on the screen and you must rebuild the original display. Its very simple - VERY addictive!

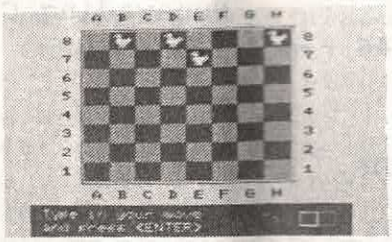
REVERSE - Your task is to arrange the ten jumbled numbers into ascending order by reversing some or all of them. Check it out - its not easy!



TS852 - MSX852 -

GRAND PRIX - You are in the drivers seat in a race to the chequered flag while avoiding slower cars. Hitting another car or two wheels off the track will slow you down and lose points.

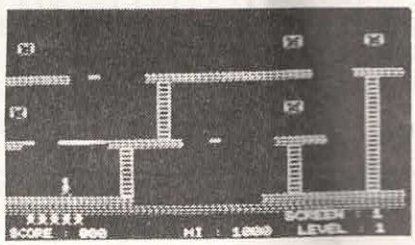
SUBSTRIKE - Drop depth charges on the enemy below. The deeper he is or the faster he moves, the more you will score.



SVPAK24 - MSXPAK24 -

FOX & GEESE - Can you as the Fox penetrate the line of advancing geese and get to the other side of the board before they corner you.

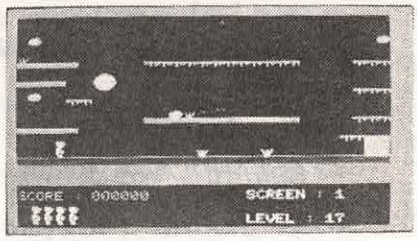
THINK - This is a form of word mastermind. You must guess the word that the computer has selected from the vast number stored in its data banks. This game will certainly make you do as the name suggests.



SVPAK7 - MSXPAK7 -

BOMB SCARE - Very addictive game in which you must climb to different platforms and defuse 5 bombs before your bonus runs out. Has 6 separate screens with the hazards of moving platforms and conveyor belts with holes in them.

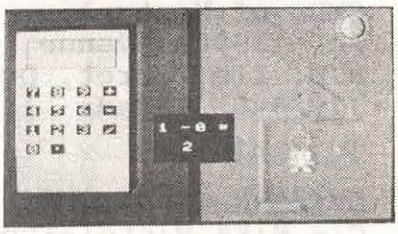
DUNGEON ADVENTURE - A text adventure where you explore the dungeon fighting monsters and gain treasures to discover the location of the great Orb.



SVPAK21 - MSXPAK21 -

GOLDRUSH - Move the miner around the different levels of the mine collecting gold and dodging high explosive charges and falling rocks. The green ledges crumble if you don't get off quickly. When you have collected all of the gold, get to the lift cage for a new screen.

PYRAMID - This is another fascinating text adventure built around the vaults of a pyramid.



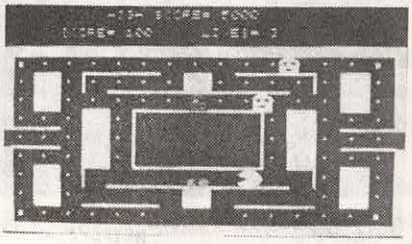
SVPAK25 - MSXPAK25 -

OZMAP - You must drive the car around Australia finding the places required as quickly as you can.

OZQUIZ - Answer general knowledge questions about Australia and her people.

MATH DUEL - Makes practicing maths fun! Has 4 levels of addition and subtraction maths.

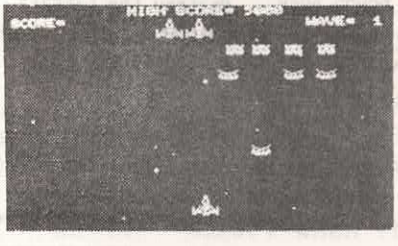
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SVPAK12 - MSXPAK12 -

MUNCH MAN - Guide your Munchman around the maze eating the dots while avoiding the deadly ghosts. If you eat one of the larger dots you will be able to eat the ghosts for a short while. Gain extra points by eating the fruit that occasionally appears on the screen.

ESCAPE FROM PLANET BETA - This is a text adventure where you are stranded on Planet Beta and must find all the necessary equipment to get off before the planet explodes. Only requires one or two-word commands - eg GET LASER.



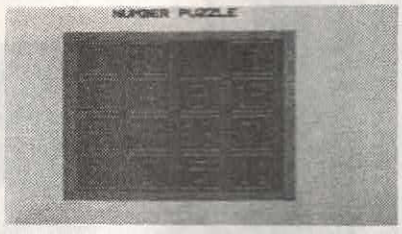
SVPAK5 - MSXPAK5 -

LUNAR LANDER - You are in command of a Lunar Excursion Module with limited fuel but full rotational ability and must land on the lunar surface. Has 8 different landscapes which get harder

GALACTIC ASSAULT - The alien invaders hover above you and one at a time dive on you. Destroy each wave before they get you. For 1 or 2 players.

ROAD RACER - Try and pass as many cars just as fast as you can without crashing.

SPRITE DEFINER - Design sprite shapes on the screen and rotate them when you have finished. Has grids for both sprite sizes.



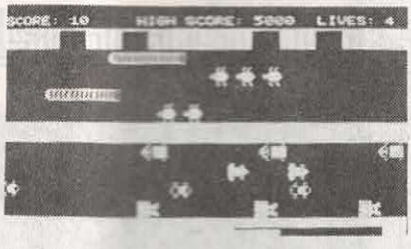
SVPAK20 - MSXPAK20 -

ARTILLERY FIRE - A two player game where each can select a firing angle and velocity to lob shells on his opponent.

NUMBER PUZZLE - Computerised version of the popular hand-held game in which you must slide numbers around into ascending order.

TOWERS OF HANOI - Move all of the discs from peg 1 to peg 3 without placing a large disc on a smaller one.

TENNIS - This is a two player game of Table Tennis.



SVPAK6 - MSXPAK6 -

HOPPER - Help Hoppy get across the road (SPLAT!) and then to cross the river (SPLOSH!) and if you are lucky he won't get eaten by the crocs (GULP!!).

SPACE WAR - Challenging game for two players in this duel between your spaceship and that of your opponent. First one to ten points wins.

OTHELLO CHALLENGER - The classic mind game where you must outflank your opponent to capture his pieces. For 1 or 2 players.

PAINTER - Explore the flexibility of your computers colours with this on-screen paint brush.

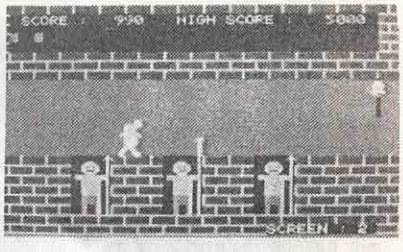


SVPAK9 - MSXPAK9 -

SPACE INVADERS - Shoot the aliens before they invade your planet. Move your ship left and right and shoot missiles at the aliens. Once they have landed the game will end. Score extra points by shooting the UFO which flies overhead.

SUB HUNT - Drop depth charges on the attacking subs and dodge the missiles they fire. You can fire two depth charges at once by firing while moving left or right. Each of the subs is marked with its value.

CHICKEN - Who cares why the chicken crossed the road, we have to find out how! Each time you make it across, the cars speed up.



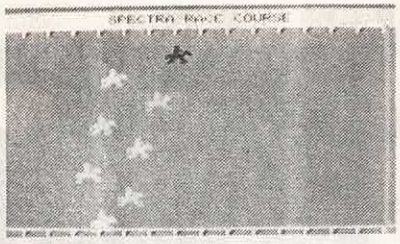
SVPAK11 - MSXPAK11 -

LUNAR RESCUE - Rescue the stranded astronauts from the moons surface one by one. Dodge the asteroids on the way down and destroy them on the way back up for more points.

SKY BASE - Blast the aliens but watch out for their kamikazi attacks. Shoot two sets and then dock with your mother ship. With this added fire-power go on to destroy the alien base.

QUASIMODO - Can you as the hunchback Quasimodo rescue the beautiful maiden from the French soldiers. Cross the screen avoiding fireballs and spears, then ring the bell to get the next screen.

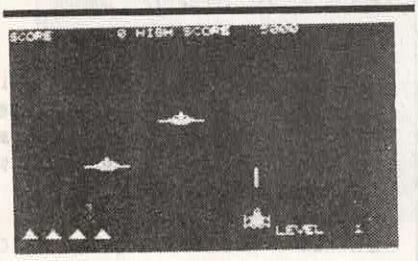
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SVPAK1 - MSXPAK1 -

SPECTRA DERBY - Place your bets and have a day at the races. Even if you lose its painless.

HANGMAN - The classic word guessing game in which you will be gradually hung for each wrong letter.

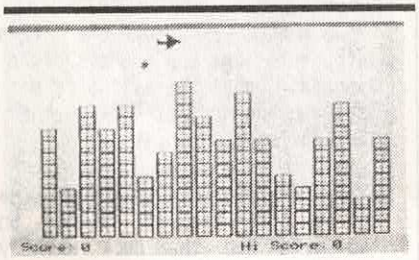


SVPAK10 - MSXPAK10 -

DEMON ATTACK - Shoot the attacking demons as they swoop overhead. You can only destroy a demon when it is fully formed. Killing ten demons gets you on to a harder set.

SEA DIVER - Dodge sea creatures as you make your way to the sea bed. Grab a gold bar and get back to the surface.

DEATH CITY - Race around the maze trying to shoot your computer opponent but be careful, he is also shooting at you.



SVPAK2 - MSXPAK2 -

SNAKE - Move the snake around to eat the diamonds, but mind you do not cross your own tail.

CITY BOMBER - You must destroy the towering buildings of the city before you can land safely.

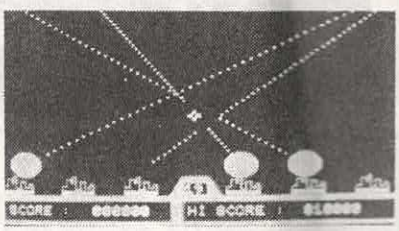


SVPAK8 - MSXPAK8 -

EXPLORER - Watch out for the Pitfalls in this game which will test your control of the joystick. Crocs and other obstacles bar your way as you wander through the jungle in search of treasures.

PARACHUTE - Catch the paratroopers in your boat as they jump from a helicopter into a shark-infested lagoon. The more you catch, the faster they jump.

ANDROID CAVERNS - Move around the caverns of this game and collect the treasure guarded by the androids. But watch out, the androids learn by their mistakes.

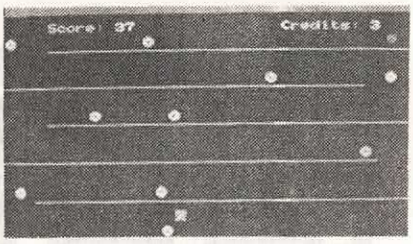


SVPAK22 - MSXPAK22 -

MISSILE COMMAND - Shoot down the incoming warheads before they destroy your cities. You get bonus cities awarded for high scores.

MASTERMIND - Your task is to match the same hidden four-colour combination that the computer has chosen.

LEAKY ROOF - Catch the raindrops leaking through the roof and empty the bucket out of the window before you drown.

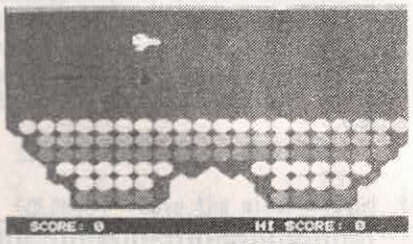


SVPAK4 - MSXPAK4 -

SPECTRA KONG - Win the lady's heart by getting to the top of the screen - but watch out for falling barrels.

TENNIS - A colour version of the original black and white TV game.

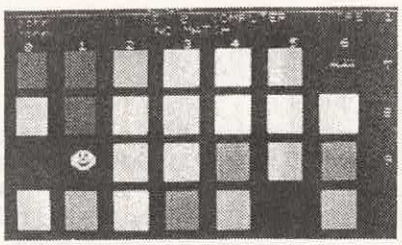
MUSICAL KEYBOARD - This program turns your computer keyboard into a musical keyboard and allows you to select note length and octave.



SVPAK3 - MSXPAK3 -

BOMBS AWAY - Destroy the targets in the ravine below. Be warned - it gets harder the more targets you hit.

BUG CATCHER - Two players move around the garden and the one with the most bugs caught - wins. Great for younger children.



SVEP1 - MSXEP1 -

COPY CAT - The computer will play a set of notes and you must play them back in sequence. There are two levels of play - one note added to the same sequence or a new sequence each time.

MEMORY - A 1 or 2 player game of memory exercise. Uncover two squares on the screen and if the objects revealed match you get another chance

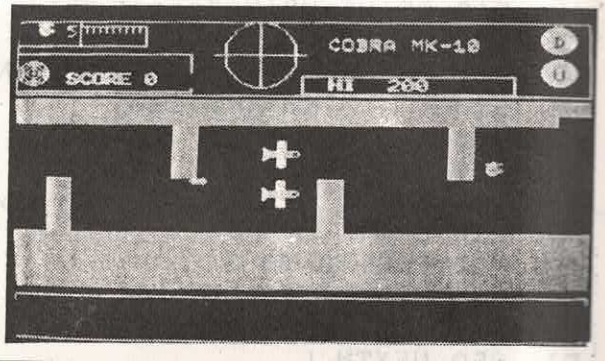
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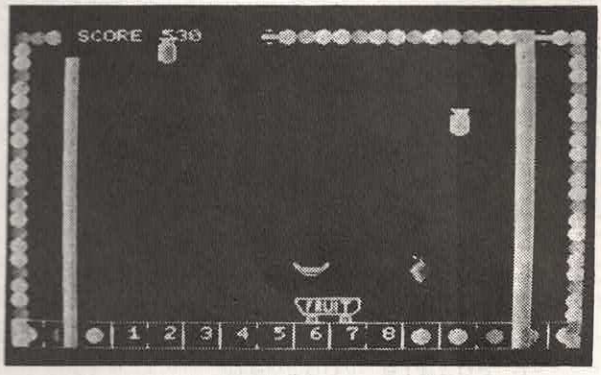
SVPAK26 - MSXPAK26 -

JUMPING COWBOY - A 'platform' game of nine screens where you must collect all the opals to get the next screen.

SUPER SMASHER - Move the bat to smash the ball through the two brick walls. Has three levels of difficulty.

FRUIT CATCHER - Catch falling fruit in your cart. As you catch more, the types of fruit will change and fall at a faster rate.





SVPAK27 - MSXPAK27 -

MIRKWOOD FOREST - Graphics adventure where you must escape from the forest by finding four keys, eating food to maintain energy, while avoiding the spiders. This one will really test your adventure gaming ability.

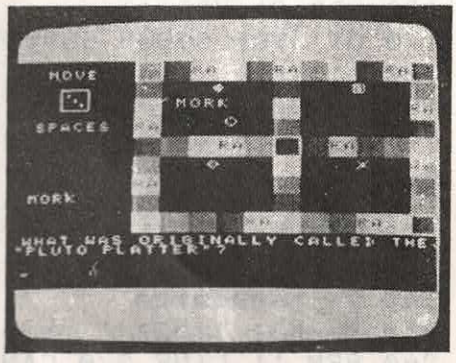
CHOPPER PILOT - The fourteen levels of this game make it very compelling. Keeping clear of canyon walls, fly your helicopter to do battle with enemy fighters, bombers, missiles and UFOs. Great skill required.

UFO (MSX ONLY)

In this machine code game, you are under attack from UFOs. Watch out for the alien unit moving across the bottom of the screen - this will fire 'smart bombs'. Watch out for the meteor shower. Refuel when the truck appears on screen. Requires 16K RAM, has smooth scrolling, fast response with good sound effects.

METEOR SWARM (SV ONLY)

Another machine code game where your ship is caught in a meteor belt and you must blast your way out. Other ships are caught in the belt and will shoot at you if you do not destroy them first. Comes with bonus game - Birds of Orion.



SPECTRA-TRIVIA (SV 32K)
COMPU-TRIVIA (MSX)

The famous trivia game now on your computer. These are trivia quiz games for up to four players. The computer rolls the dice and each player moves in turn. At certain points you will be asked questions. There is provision for you to create your own question and answer file.

PLEASE SPECIFY SV318/328 OR MSX WITH ORDER.

PUZZLE

Here is a good little MSX puzzle program from GARY GREEN of QLD.

```
EL  10 CLS
AC  20 '   by Gary green
DC  30 '        1988
BE  40 KEYOFF
GA  50 KEY1,"screen0"+CHR$(13)
GO  60 KEY2,"RENUM10"+CHR$(13)
FB  70 CLS
HL  80 PRINT"This is like the numbe
    r Puzzle you can"
BM  90 PRINT"buy in a shop, where y
    ou mix up all    of the numbers
     and then try to get themin seq
    uence."
BM  100 PRINT:PRINT"This game is on
     the same principle     except
    the computer mixes up letters o
    fthe alphabet and all you have
    to do is get them in the correc
    t order."
EB  110 PRINT:PRINT"A good challeng
    e on the mind."
BB  120 PRINT:PRINT:INPUT "    PRES
    S ENTER TO CONTINE";ZZ
DG  130 DEFINT A-Z
BG  140 COLOR 10,1,1:HS=1000
CC  150 SCREEN1,1,0
BE  160 VDP(6)=VDP(4)
BJ  170 ON STOP GOSUB 560:STOP ON
GE  180 ''''''''  start
AC  190 GOSUB 570:'  initialise
GF  200 GOSUB 600:'  initialise arr
    ay
EB  210 GOSUB 680:'  draw array
HN  220 GOSUB 1180:'  draw sprites
AG  230 GOSUB 1100:'  draw screen
HI  240 '  randomize
AF  250 T=RND(-TIME)
AL  260 REM '' LINE 270 CAN MIX UP
    THE NUMBERS FROM 50 TO 500 IT'S
     UP TO YOU HOW HARD YOU MAKE IT
    .
CD  270 FORZ=0 TO 70
FJ  280 RN=INT(RND(1)*8)+1
EO  290 ON RNGOSUB 750,780,800,830,
    850,880,900
IA  300 GOSUB 1180:'  draw sprites
DF  310 NEXTZ
BC  320 FORL=0 TO 4
JP  330 LOCATE10,0:PRINT"READY":BEE
    P
BD  340 LOCATE 0,19:PRINT SPC(40)
DE  350 FORT=1 TO 500:NEXTT
DF  360 LOCATE10,0:PRINT"     "
DG  370 FORT=1 TO 500:NEXTT
CO  380 NEXTL
CK  390 ''''''''''''''''''''''''
BA  400 '       main loop
AP  410 GOSUB 470:'  call joystick
HP  420 GOSUB 1180:'  draw sprites
GB  430 GOSUB 950:'  check for win
AI  440 IF F=1 THEN GOTO 1040
AC  450 GOTO 400
CO  460 ''''''''''''''''''''''''
AJ  470 D=STICK(0)
CC  480 IF D=0 THEN 470
AO  490 F=0
DK  500 ON D GOSUB 750,780,800,830,
    850,880,900
AJ  510 LOCATE12,15
BK  520 SC=SC+ABS(F=1)
AH  530 F=0
AD  540 PRINT USING"####";SC
CA  550 RETURN
DD  560 COLOR 15,4,4:SCREEN0:END
BA  570 '      initialise
AG  580 H=3:V=3:SC=0
BM  590 RETURN
JB  600 '       initialise array
AB  610 C=0
AO  620 FORL=0 TO 3
AO  630 FORM=0 TO 3
LM  640 CH(L,M)=65+C:C=C+1
EJ  650 NEXTM,L
FK  660 CH(3,3)=32
FN  670 '              draw array
BE  680 FORL=0 TO 3
BE  690 FORM=0 TO 3
AP  700 LOCATE15+L,5+M:PRINTCHR$(CH
    (M,L))
EE  710 NEXTM,L
CF  720 RETURN
DE  730 ''''''''''''''''''''''''
BA  740 '      joystick
AK  750 IF V=3 THEN RETURN
MC  760 SWAP CH(V,H),CH(V+1,H)
EF  770 V=V+1:F=1
BP  780 RETURN
CO  790 ''''''''''''''''''''
AN  800 IF H=0 THEN RETURN
MG  810 SWAP CH(V,H),CH(V,H-1)
CB  820 H=H-1:F=1
CF  830 RETURN
DE  840 ''''''''''''''''''''
AG  850 IF V=0 THEN RETURN
MB  860 SWAP CH(V,H),CH(V-1,H)
EC  870 V=V-1:F=1
CA  880 RETURN
CP  890 ''''''''''''''''''''
AL  900 IF H=3 THEN RETURN
MJ  910 SWAP CH(V,H),CH(V,H+1)
CC  920 H=H+1:F=1
CG  930 RETURN
```

```
AO  940 ''''''''''''''''''''
II  950 '  check for win
CG  960 A$="":F=0
BA  970 FORL=0 TO 3
BA  980 FORM=0 TO 3
HN  990 A$=A$+CHR$(CH(L,M))
HL 1000 NEXTM,L
FI 1010 IF A$="ABCDEFGHIJKLMNO "TH
    EN BEEP:F=1
BD 1020 RETURN
CL 1030 ''''''''''''''''''''
IC 1040 IF SC<HS THEN HS=SC:BEEP
EE 1050 LOCATE12,16
FG 1060 PRINT USING"####";HS
BI 1070 FORT=1 TO 1000:NEXTT
CO 1080 GOTO 190
DB 1090 ''''''''''''''''''''''''
DH 1100 '     draw screen
AP 1110 LOCATE0,15
AP 1120 PRINT "Move Number>"
FG 1130 PRINT "Best Yet   >"
LC 1140 PRINT:PRINT:PRINT"MIXING T
    HINGS UP PLEASE WAIT"
BF 1150 RETURN
DH 1160 HS=1000
BH 1170 RETURN
DL 1180 '     draw sprites
DP 1190 C=0
EB 1200 FORL=0 TO 3
EB 1210 FORM=0 TO 3
ID 1220 PUTSPRITE C,(40+(L*16),40+
    (M*16)),2,CH(M,L)
FB 1230 C=C+1
HN 1240 NEXTM,L
BE 1250 RETURN

      THE END
```


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MENU

This program, from Bruce Goodman, is for use on the SVI computers, with disk drives. You place a copy on each disk and set it up to run on boot-up. ie use the IPL command % = LEFT+0, ñ = LEFT+E, ij = LEFT+Z ó = LEFT+C, ▒ = RIGHT+X, ▒ = RIGHT +Y, ▒ = RIGHT+O

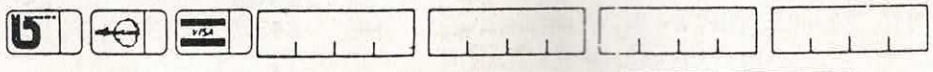
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BF  10 CLEAR 1000:GOSUB 930
FJ  20 OUT&H88,15:OUT&H8C,255:POKE&
    HFE38,1
AM  30 SCREEN0,0:LOCATE,,0:COLOR7,1
    :CLS
LK  40 DEFSTRA-H:DEFINTI-Z:GOSUB300
    :DIMF[35,1]:FIELD#0,128ASD1,128
    ASD2:P=0:YR=12
BD  50 FORI=1TO13:C=DSKI$(1,20,I):D
    =D1:GOSUB170:D=D2:GOSUB170:NEXT
    I
EJ  60 IFP=0THENGOSUB310:LOCATE,,1:
    END
DM  70 IFP=1THENP=0:GOTO90
KE  80 P=3*((P-1)\6)+1
BA  90 GOSUB410:GOSUB430:GOSUB120
AB  100 GOSUB550:GOSUB730:GOTO100
CA  110 '<<<<<< Flashing Message >>
    >>>>>>
AM  120 E="Use Joystick to select P
    rogram....":GOTO150
DA  130 E="ASCII File...Attempt Run
     (Y/N) ?":GOTO150
FK  140 E="Screen File...Can't Run"
    :GOTO150
CG  150 BEEP:GOSUB910:ONINTERVAL=50
    GOSUB480:INTERVALON:RETURN
CJ  160 '<<<<<<< Find & Print Files
     >>>>>>
AG  170 FORJ=0TO7:E=MID$(D,(16*J)+1
    ,16)
AC  180 IFLEFT$(E,1)=CHR$(255)THENR
    ETURN60
BB  190 IFLEFT$(E,1)=CHR$(0)THEN280
CL  200 Z=ASC(MID$(E,10,1))
AI  210 IFZ=160ORZ=224ORZ=176THENG=
    "#":GOTO250
CO  220 IFZ=0ORZ=16ORZ=64THENG=" ":
    GOTO250
CE  230 IFZ=1ORZ=17ORZ=65THENG="*":
    GOTO250
AO  240 G="."
DN  250 F[P,1]=MID$(E,1,6)+G+MID$(E
    ,7,3)
EM  260 IF(P+3)MOD6=0THENLOCATE0,5:
    PRINTCHR$(27)+"M":YR=YR-1
NO  270 GOSUB410:LOCATEX,Y:PRINTF[P
    ,1]:P=P+1
AD  280 NEXTJ:RETURN
CH  290 '<<<<< Print Screen Heading
     >>>>>
DK  300 A="DISK MENU":Y1=0:GOTO 320
BJ  310 A="This disk is Empty !!":Y
    1=11
FO  320 A=CHR$(211)+CHR$(210)+" "+A
    +" "+CHR$(210)+CHR$(211)
EH  330 X1=20-LEN(A)/2
AK  340 LOCATE X1,Y1
DE  350 PRINT"%";STRING$(LEN(A)-2,"
    ▒");"ñ"
BM  360 LOCATE X1,Y1+1:PRINT A
BE  370 LOCATE X1,Y1+2
DO  380 PRINT"ij";STRING$(LEN(A)-2,"
    ▒");"ó"
BK  390 RETURN
HP  400 '▒▒▒▒▒▒▒ Using P Find X,Y ▒
    ▒▒▒▒▒▒▒
BJ  410  Y=YR+P\3:X=PMOD3+2+(PMOD3)
    *12:RETURN
AD  420 '▒▒▒▒▒▒ Turn On File P ▒▒▒▒
    ▒▒▒▒▒▒
GG  430 LOCATE X-1,Y
GB  440 PRINT CHR$(27)+"p";"▒";
IF  450 PRINT F[P,1];CHR$(27)+"q"
BO  460 RETURN
JB  470 '▒▒▒▒▒▒▒▒ Message ▒▒▒▒▒▒▒▒
    ▒▒▒▒
GK  480  LOCATE 19-LEN(E)/2,21
DK  490 IF Z<>0 THEN 520
JM  500  PRINT CHR$(27)+"p";"▒";E;C
    HR$(27)+"q";
BA  510  Z=1:RETURN
KJ  520 PRINT " ";E;CHR$(27)+"q";
AP  530 Z=0:RETURN
AF  540 '▒▒▒▒▒▒▒▒ STICK RESPOND ▒▒▒▒
    ▒▒▒▒▒
DD  550 ST=STICK(0)+STICK(1):IFST=0
    THEN710ELSEBEEP
FM  560 INTERVALSTOP:P1=-1
BH  570 IF ST=1 THEN P1=P-3
BI  580 IF ST=3 THEN P1=P+1
BL  590 IF ST=5 THEN P1=P+3
CD  600 IF ST=7 THEN P1=P-1
CF  610 IF P1<0 OR P1>35 THEN 690
CE  620 IF F[P1,1]="" THEN 690
CO  630 GOSUB 410
GF  640 LOCATE X-1,Y
DM  650 PRINT" ";F[P,1]
AC  660 P=P1
CK  670 GOSUB 410
DE  680 FOR I=1 TO 50:NEXT I
CG  690 GOSUB 430
DL  700 INTERVAL ON
CG  710 RETURN
IH  720 '▒▒▒▒▒ Trigger Response ▒▒▒▒
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BP  730 TR=STRIG(0) + STRIG(1)
AO  740 IF NOT TR THEN RETURN
DJ  750 INTERVAL STOP
JF  760 IF MID$(F[P,1],7,1)="."THEN
    GOSUB890:RUN"1:"+F[P,1]
FA  770 LOCATE 0,21:PRINT CHR$(27)+
    "l"
FK  780 IF MID$(F[P,1],7,1)<>" "THE
    N 830
AM  790 BEEP:GOSUB 130
AG  800 B=INKEY$:IFB=""THEN800
LP  810 IFB="Y"THENCLS:MID$(F[P,1],
    7,1)=".":GOSUB890:RUN"1:"+F[P,1
    ]
CP  820 IFB="N"THENGOSUB120:RETURNE
    LSE800
AJ  830 IF MID$(F[P,1],7,1)<>"#"THE
    N870
BD  840 BEEP:GOSUB140
BC  850 FOR I=1TO8000:NEXT I:BEEP
AB  860 GOSUB 120:RETURN
BO  870 CLS:MID$(F[P,1],7,1)=".":GO
    SUB890:BLOAD"1:"+F[P,1],R
JJ  880 '◆◆◆◆◆◆◆ Clear keyboard ◆◆◆
    ◆◆◆◆◆
AL  890 B=INKEY$:IFB=""THENRETURNEL
    SE890
IE  900 '◆◆◆◆◆◆◆ Clear Line 21 ◆◆◆◆
    ◆◆◆◆◆
HC  910 LOCATE0,21:PRINTCHR$(27)+"l
    ":RETURN
IF  920 '◆◆◆◆◆◆◆◆ Begin Module ◆◆◆◆
    ◆◆◆◆◆
HE  930 ZZ$="START.TST"
AK  940 GOSUB 950:GOTO 1450
CE  950 FORI=1TO10:KEY(I)ON:NEXTI
BN  960 ON KEY GOSUB 1120,1160,1310
    ,970,1190:RETURN
BI  970 INTERVAL OFF
DH  980 COLOR1,1,1:SCREEN0,0
BO  990 CLS:LOCATE 10,10,0
CI  1000 PRINT "Save as ";ZZ$
FJ  1010 LOCATE16,12 : PRINT "Y or
    N"
AB  1020 INTERVAL ON
CI  1030 ON INTERVAL=25 GOSUB 1080
    : Z$=""
HG  1040 Z$=INKEY$:IF Z$="" THEN 10
    40
CO  1050 IF Z$="Y"ORZ$="y"THENSAVE"
    1:"+ZZ$:      INTERVALOFF:COLOR
    10,1:CLS:RUN
HP  1060 IF Z$="N" OR Z$="n" THEN
                    INTERVAL OFF:RUN
DC  1070 BEEP:GOTO 1040
BF  1080 COLOR10:ON INTERVAL=25 GOS
    UB 1100
CG  1090 BEEP : RETURN
DP  1100 COLOR1:ON INTERVAL=25 GOSU
    B 1080
BB  1110 RETURN
DJ  1120 GOSUB 1220
EN  1130 INTERVAL OFF
FB  1140 COLOR11,1,1
DJ  1150 SCREEN0,1:LOCATE,,1:LIST -
    930'
FA  1160 INTERVAL OFF
AP  1170 COLOR1,1,1
GF  1180 SCREEN0,0:LOCATE,,0:RUN
BB  1190 COLOR1,1,1
EN  1200 SCREEN0,1:LOCATE,,1
FB  1210 INTERVALOFF:GOSUB1220:COLO
    R11:END
EN  1220 CLOSE:KEY 1,"CLRLIN"+CHR$(
    21)
GM  1230 KEY 2," AUTO":KEY 3," LIST
    "
AJ  1240 KEY 4,CHR$(21)+"CLS:LIST"+
    CHR$(13)
AA  1250 KEY 5,CHR$(21)+" RUN"+CHR$
    (13)
EB  1260 KEY 6," RUN"+CHR$(34)+"1:"
JN  1270 KEY 7,"SAVE"+CHR$(34)+"1:"
FA  1280 KEY 8,"LOCATE,,1"+CHR$(13)
    +CHR$(30)+CHR$(21)+CHR$(30)+CHR
    $(21)+CHR$(30)+CHR$(21)
GA  1290 KEY 9,"COLOR 2,1,1"+CHR$(1
    3)
EG  1300 KEY 10," FILES"+CHR$(13):R
    ETURN
EJ  1310 INTERVAL OFF
EK  1320 COLOR 13,1
EJ  1330 SCREEN 0,0
AE  1340 LOCATE10,5,0
HO  1350 PRINT "Total Memory = 2119
    1"
DM  1360 LOCATE11,10:PRINT "Memory
    Left =";
DB  1370 PRINT USING "######";FRE(0
    )
EN  1380 LOCATE 11,15:PRINT "Memory
     Used =";
DG  1390 PRINT USING "######";21191
    -FRE(0)
AJ  1400 GOTO 1400
BB  1410 ON ERROR GOTO 1420 : GOTO
    1460
HE  1420 COLOR10,1:SCREEN0,1:LOCATE
    9,9,1
IO  1430 PRINT "Error No";ERR;"in L
    ine";ERL
EC  1440 GOSUB1220:PRINT:LIST.:END
AK  1450 GOSUB1220:STOP ON : ON STO
    P GOSUB 1190
BD  1460 RETURN
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